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RESEARCH ARTICLE

De-Randomization of MAC Addresses Using Fingerprints and RSSI With ML for Wi-Fi Analytics

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ABSTRACT *Media Access Control* (MAC) address randomization causes significant distortion and data loss in Wi-Fi analytics systems, becoming a real challenge for building services based on tracking, location, and presence data. This study aims to mitigate this problem by combining two key points: the construction of a quasi-unique, stable, reliable, and anonymous identifier for non-connected Wi-Fi devices, and the inability of Wi-Fi devices to deliberately change the physical conditions of the connection. We propose a new system that builds identifiers based on the capabilities and information elements announced within the probe request management frames, and consequently applies unsupervised machine learning techniques in the multidimensional *Received Signal Strength Indicator* (RSSI) space. Experimental tests in a real-world environment were conducted, and the results of this extensive field study demonstrated that the proposed system achieves high accuracy in identifying and tracking non-connected Wi-Fi devices in these challenging scenarios, even in the presence of MAC randomization. Our findings suggest that the proposed system has a significant potential for enhancing building services that rely on Wi-Fi data.

INDEX TERMS Data processing, IEEE 802.11 networks, machine learning, MAC randomization, passive sniffing, system validation, wireless access points, wireless fidelity (Wi-Fi), wireless LAN, Wi-Fi analytics.

I. INTRODUCTION

In the realm of Wi-Fi technology [1], rapid expansion and ubiquity have led to more than 18 billion devices globally, with projections showing continual growth [2], [3]. This widespread adoption has underscored Wi-Fi's critical role in enabling connectivity and facilitating a plethora of applications, from Internet access to location-based services [4]. However, the advent of *Media Access Control* (MAC) address randomization, initiated by privacy concerns and implemented in operating systems in the mid-2010s, has introduced significant challenges in

Wi-Fi analytics. Since iOS 8 and Android 6, released in 2014 and 2015 [5], respectively, devices no longer show their real MAC addresses as the source address in probe request frames, but they use a locally administrated address [6], which is a fake address that changes continuously and randomly when devices are not associated with the Wi-Fi network. Traditional methods, designed to track and count devices based on their MAC addresses, are now facing substantial hurdles because these MAC addresses are no longer static or reliable identifiers [7], [8], [9], [10].

MAC randomization is currently performed by more than 80% of devices [11] when they are not associated. New methods must be employed to counteract the decline caused

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by MAC randomization and ensure the reliable acquisition of actionable data. Various attempts have been made to solve the problem of MAC randomization, such as the use of *Service Set Identifiers* (SSIDs), sequence numbers, certain non-random bits of the MAC, and stimulation with karma or honeypot attacks. All these techniques are considered vulnerabilities and have stopped working, generally, with up-to-date smartphones and tablets. These approaches and the reasons for their failure are discussed in Section III. Our system implements a method embedded in Access Points (APs) that generates fingerprints based on the information elements contained in the 5-GHz probe request frames. More importantly, this study extends and improves the results of [12] by performing an entire data pre-processing stage, followed by *Machine Learning* (ML) final clustering, which improves the accuracy of the first stage. The reason for this enhancement is that the fingerprint generation stage has the corner case of delivering the same fingerprint for different devices with the same vendor brand, model, firmware version, and operating system (OS) version. When that happens, no information element in the 802.11 frames can help the system to differentiate both devices, leading to the so-called fingerprint collision. Fortunately, one of the main contributions of this study is post-processing, which aims to minimize this problem. The *Received Signal Strength Indicator* (RSSI) has always been a key point for location and positioning in Wi-Fi networks [13], [14], [15], [16], [17], [18], but not to differentiate one device from another, where the instability of the RSSI becomes less relevant. In the proposed system, a post-processing blind RSSI clustering stage was implemented to overcome the problem of fingerprint collisions.

Finally, the efficacy of the proposed system was tested by conducting an extensive experiment in a real-world scenario. A Wi-Fi counting system was deployed on a university campus for two days. This unique configuration is part of a commercial entity, reflecting a blend of academic insights and industrial rigor. This setup ensured that the results of this system were not only theoretically sound but also industrially viable and applicable. Without collaboration, students, professors, and workers stayed and passed through the vicinity of the sensor with non-associated devices. They were counted by hand, as a ground truth, and by using the standard “only real MACs” approach in the industry [9], [19], [20], [21]. Both counting systems were compared with the results of our solution. The results show that MAC randomization is effectively mitigated, and Wi-Fi analytics based on these techniques show a considerable accuracy of almost 90% in real-time counting and 98% in overall experiment counting. This is an improvement over the industry standard used by Meraki and other classical Wi-Fi analytics systems, which attains 22% accuracy in both metrics.

Hence, the main contributions of this paper are:

- A more detailed explanation of the probe request fingerprinting algorithm first introduced in [12].
- The combination of fingerprinting techniques by puncturing information elements with unsupervised clustering in a multidimensional signal strength space to separate similar devices.
- Extensive field testing of the accuracy of this system against the ground truth of counting by hand and the standard industry Wi-Fi counting system, in a multi-day experiment with hundreds of real devices and passive non-cooperating users.

This paper is structured as follows: In Section II, an in-depth explanation of active scanning, probe request frames, and MAC randomization is presented, providing the basis for understanding the addressed problem properly. Section III summarizes state-of-the-art solutions to the problem and clarifies the novelty of this study. Section IV details each stage of the proposed de-randomization system. Experimental tests were conducted, and the results are presented in Section V. Finally, the conclusions are presented in Section VI.

II. PRELIMINARIES

A. 802.11 ACTIVE SCANNING: PROBE REQUEST ANALYSIS

The IEEE 802.11 Standard [1], widely known under the brand Wi-Fi, describes a set of protocols intended to provide wireless connectivity for fixed, portable, and moving wireless user stations (STAs) within a local area, where APs act as nodes that enable access to the distribution system. For an STA to be able to communicate with an AP, its MAC address must first be known, along with other parameters such as the logical name of the network, supported rates, and protocol capabilities. The standard defines two different methods that allow the STA can obtain this information: passive and active scanning. Each of these procedures involves different management frames. During passive scanning, the STA listens to the beacon frames of the APs, which are periodically broadcasted on every channel. However, this is not an efficient method for an STA to discover all APs available in all the channels. Therefore, STAs can actively scan their surroundings by broadcasting management frames, known as probe requests. All APs that receive the request generally send back a probe response frame containing similar information to the beacon frame [4].

Probe request frames can contain the SSID of the specific network that the STA is looking for or can be configured to search for any SSID. Moreover, STAs generally try to ensure the reception of frames by sending the same probe request many times within a short period in the order of milliseconds. These are known as probe request bursts. The number of frames per burst, as well as the cadence at which bursts are sent, from seconds to minutes, depends on the implementation by the manufacturers, as the standard does not specify this detail. Factors such as the OS version, remaining battery, brand and model of the device, and proximity to APs in the vicinity may affect how active scanning is performed.

Active scanning involves two key aspects. First, as STA is the one searching for APs, the probe request frame is the

only 802.11 frame that devices transmit when they are not connected to any 802.11 network and even have no intention of doing so. Therefore, capturing and analyzing these frames, known as passive sniffing, is the most important approach for obtaining key analytics of non-associated devices. Second, this frame also contains some payload parameters to provide prior information to the AP regarding the capabilities of the STA. These parameters are called Information Elements (IEs), and many of them are optional fields whose utilization depends on the configuration of the device as well as the versions of the IEEE 802.11 Standard it supports. The general format of the probe request frame is listed in Table 1. Optional IEs are marked by *. The source MAC address, which corresponds to the MAC address of the STA, is indicated in bold. As mentioned in the previous section, manufacturers recognized that transmitting the MAC address of the device in probe request frames poses a privacy risk to users. Consequently, they implemented a procedure known as MAC randomization during active scanning, to hide the actual MAC address of the STA. Although using a Virtual Private Network (VPN) is often the first measure adopted to enhance privacy, it's important to note that VPNs operate at layer 3 of the OSI model or above, primarily to encrypt traffic and secure data transmissions. However, because VPNs function at a higher layer, they do not impact the broadcasting of probe request frames, which occur at the lower MAC layer (layer 2) and are independent of the network layer activities. Thus, although VPNs provide significant privacy benefits, they do not alter or affect the capture and analysis of 802.11 frames.

B. MAC RANDOMIZATION

MAC randomization is a process in which STAs generate temporary virtual MAC addresses during the active scanning phase, aiming to keep the real hardware MAC address concealed to ensure anonymity. The device changes the MAC address shown in the header of the probe request frames at an indeterminate cadence. The OS of the device manages this process, and it varies with each OS implementing randomization method, as there are no standard algorithms. Some may change only the last three bytes, keeping the same Organization Unique Identifier (OUI), whereas others may change all possible bits.

Fortunately, the IEEE defines guidelines for the use of MAC addresses. In particular, of the 48 bits of the MAC address, the 2nd least significant bit of octet 0 (U/L bit) indicates universal (U/L=0) or local (U/L=1) administration of the address, that is, whether the address is real or random, respectively [6]. Hence, by checking this bit, it is possible to separate the analysis of devices with non-random MAC addresses, the real static identifier of the chipset, and those with a continuously changing random MAC. Devices that send their globally unique MAC addresses exhibit a single MAC in all their frames, whereas devices with randomized MAC addresses have dozens or even hundreds of different MAC addresses per hour. Its structure is shown in Fig. 1.

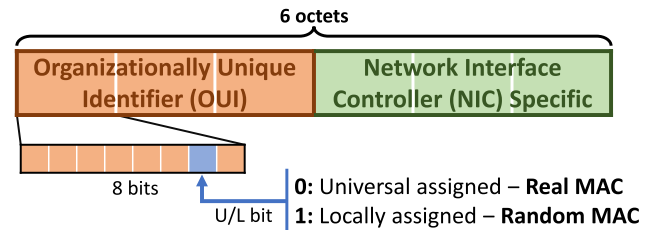


FIGURE 1. General structure of the MAC address, where the 2nd least significant bit of the first octet is the (U/L) bit, that indicates whether an address is real or random, according to the 802.11 standard [6].

The core issue in Wi-Fi analytics is the randomization of MAC addresses in Wi-Fi devices that are not connected to a network. This phenomenon leads to ambiguity and substantial inaccuracies in analytics systems that rely on MAC addresses to identify the devices. Several devices may be initially tracked when looking at different MAC addresses, whereas in reality, it is only one device that is randomizing its address. Focusing this issue on smartphones, it is estimated that 80-90% of modern smartphones exhibit this behavior. Those that do not exhibit this behavior typically have older versions of the OS [11]. Thus, it is easy to anticipate that the percentage of devices that randomize their MACs will increase to 100% in the next few years as users upgrade their smartphones.

Furthermore, the pattern of randomization cannot be straightforwardly deduced or rectified because APs acting as external observers cannot discern which MAC addresses are associated with the same device based solely on this identifier. Consequently, the challenge for Wi-Fi analytics systems is evident: relying on MAC addresses as a dependable means of identifying devices significantly compromises the precision of counting, tracking, and segmenting devices owing to the considerable distortion introduced by address randomization. The first solution implemented by the industry was to discard every frame in which the MAC address was randomized. Initially, it served as a workaround, but as more smartphones were updated, MAC randomization became standard, and is currently carried out by more than 80% of the devices [11]. Hence, the results of analytics with discarded frames are highly biased, and new methods must be employed to counteract the deceit of MAC randomization, ensuring the acquisition of dependable data for executing consistent business strategies.

III. DE-RANDOMIZATION TECHNIQUES: PAST AND PRESENT

Numerous proposals have emerged in recent years to correct or amend, to some extent, the distortion produced by the MAC randomization processes present today in almost all mobile devices. In this section, we analyze the evolution of these techniques and their countermeasures.

It must be considered that the probe requests initially started randomizing only the Network Interface Controller-specific part of the MAC addresses, that is, the most significant 24 bits (cf. Fig. 1). Thus, the OUI remained

TABLE 1. Anatomy of the probe request frame.

MAC HEADER	Frame Control Field: 0x0004	
	Duration	
	Destination MAC address: FF:FF:FF:FF:FF:FF	
	Source MAC address	
	Basic Service Set ID: FF:FF:FF:FF:FF:FF	
	Sequence control number	
FRAME BODY	Service Set ID (SSID) parameters	Tag number: 0x00
		SSID length
		SSID
	Supported rates	Tag number: 0x01
		Tag length
		Supported rates
	Extended supported rates*	Tag number: 0x32
		Tag length
		Extended supported rates
	Direct Sequence parameter set*	Tag number: 0x03
		Tag length: 1
		Current channel
	High Throughput (HT) capabilities*	Tag number: 0x2D
		Tag length: 26
		HT capabilities info
		A-MPDU parameters
		Rx supported MCS set
		HT extended capabilities
		TxBF capabilities
	Extended capabilities*	Antenna selection capabilities
		Tag number: 0x7F
		Tag length: 8-10
	Very High Throughput (VHT) capabilities*	Extended capabilities
		Tag number: 0xBF
		Tag length: 12
		VHT capabilities info
		VHT supported MCS set
		Rx MCS Map
	Vendor-Specific*	Tx MCS Map
		Tag number: 0xDD
		Tag length
	Fast Initial Link Setup request parameters*	Payload
		Ext tag number: 0x02
		Ext tag length: 2
	High Efficiency (HE) capabilities*	Ext tag data
		Ext tag number: 0x23
		Ext tag length: 21-29
		HE MAC capabilities
		HE PHY capabilities
		Supported HE-MCS and NSS Set
		PPE Thresholds

fixed and could be matched with specific manufacturers of emitting devices [22]. Indeed, manufacturers, including those of chipsets, final devices, and even OS, purchase segments of the MAC's vendor-specific parts corpus to assign unique physical addresses to their devices. However, in recent years, manufacturers have decided to randomize as much MAC

information as possible —46 out of 48 bits—, to prevent matching with the vendor and to remove one of the most important fields for identifying the emitting device.

First, the main approach in the scientific community was to show that the randomization technique was not perfect because its implementation lacked robustness. In fact, some methods based on active sniffing succeeded in reverting the randomization and obtaining the global MAC address. These were based on packet-manipulation attacks that manage to confuse the devices to make them broadcast their global MAC or an identifier linked to it. Vanhoef et al. [23] analyzed WiFi-Protected-Setup (WPS)-based attacks, which leveraged a parameter called Universally Unique Identifier-Enrollee (UUID-E) provided by the devices. This parameter is associated with the global MAC address of the device; hence, the devices could be easily traced. Chatzisofroniou et al. reviewed in [27] and [28] other approaches that exploited vulnerabilities in the 802.11 association and network discovery protocols, such as beacon attacks, karma attacks, and Hotspot 2.0 honeypots, which try to stimulate non-associated devices to reveal their real MAC. Although they did indeed work in the early years of MAC randomization, since iOS 10 and Android 11 these attacks have almost been fixed and no longer have the impact they used to on newer smartphones. Moreover, these attacks intended to highlight the flaws in the implementation of MAC randomization and their usage to perform Wi-Fi analytics would violate the privacy of users.

However, some de-randomization techniques have focused on passive sniffing point of view. These methods attempt to distinguish different devices by analyzing the received probe request frames. One widely used technique for constructing device identifiers, even when the entire MAC address is randomized, was the use of the sequence number. This is the last field in the MAC header represented in Table 1 and is a 2-byte field that increases with each management frame sent. Initially, probe requests from the same device had increasing sequence numbers until they reached 2048 or 4096 (depending on the manufacturer) and then reset to zero, only to continue increasing at the same rate. For example, three consecutive probe request frames from the same device could display random MAC addresses, but consecutive sequence numbers, such as 56, 57, and 58. This made it easier to match these frames as belonging to the same device. Tan et al. used this principle as a metric to evaluate the similarity between different probes in their probe association network [25]. Unfortunately, nowadays most modern devices also randomize the sequence number [29], rendering it useless for tasks such as the unique identification of the emitter.

Numerous efforts have also been made to determine the correlation between SSIDs [30], [31], [32]. An SSID is the name that the network administrator assigns to their WiFi network so that nearby devices can find it. It is well known that, historically, probe requests included a field with some of the SSIDs known in the past by the device in question, as their primary purpose was to inquire if known

TABLE 2. Summary of the main contributions of the literature on de-randomization techniques.

Proposals	Obsolete approaches				Fingerprinting				Post-processing	
	Leaks	WPS	SSID	Sequence Number	IE IDs	IE length	IE content	Residual MAC info	RSSI	Temporal
[23]	X	X	X	X	X	X	X			
[24]				X	X					
[25]				X	X	X			X	
[26]				X	X	X			X	X
This paper					X	X	X	X	X	X

networks were available to offer Internet access services. This was later considered as a significant vulnerability, which is why devices began emitting frames without searching for any specific network —SSID parameter set to *Wildcard*. Although some devices still emit a small set of SSIDs, recent studies have concluded that more than 80% of them send non-targeted probe requests [33]. Therefore, SSIDs do not provide sufficient information for constructing reliable emitter identifiers.

As we have analyzed, in recent years, the possibilities for innovation in the construction of unique and stable identifiers that do not depend on the MAC from the information contained in the probe request frames have been closing off. This movement has been mainly motivated by the privacy protection actions of smartphone and operating system manufacturers, such as Apple or Android [34]. Vanhoef et al. [23] showed that IEs could be used as the main raw material when building a unique identifier or at least as the first step toward full identification. Nitti et al. compared the IEs present in each probe request frame as the first step in their de-randomization algorithm proposed in [24], ignoring the content and length of the parameter. Pronello et al. also developed a de-randomization technique shown in [35], which uses only one of all the IEs, *Supported Rates*. Uras et al. took this approach a step further by combining some temporal information such as the time between frames, with the ID and length of the present IEs, ignoring their content [26].

Following this line, this paper presents a new system that makes use of a tailored selection of IEs, their presence and content, as well as some ad hoc data for fingerprint construction, such as residual MAC address information, together with post-processing based on unsupervised multidimensional clustering, in the signal strength space, to separate even identical fingerprints that do not belong to the same physical device. Hence, this proposal avoids the use of obsolete approaches and makes the most of all the information included in the probe request frames.

Table 2 summarizes this literature review. Active sniffing approaches were filtered because of their misconduct. The most relevant proposals were compared by the elements of information used in their de-randomization techniques, highlighting the obsolete approaches. Thus, the novelty and

contribution of this study to the present state of knowledge are clarified.

It is important to highlight that these techniques aim to mitigate the impact of MAC randomization on analytics, but do not compromise or violate user privacy. These methods succeeded in distinguishing between devices while preserving their anonymity without identifying a specific device or its owner.

IV. PROPOSED SYSTEM

In this study, a new approach for identifying non-associated Wi-Fi devices that use randomized MAC addresses is presented. The system can be divided into two fundamental stages, as illustrated in Fig. 2:

- 1) **Construction of the fingerprint:** The fingerprint is a captured profile derived from observing the capabilities in the 5 GHz probe request frame of the device. It is built using the IEs of the frame that do not change from the frames of the same device, as well as all possible invariant information that can be acquired. Even so, multiple devices sharing identical capabilities will display the same fingerprint, unraveling the need for a secondary stage in the process. It is important to highlight that the construction of the fingerprint follows an edge computing philosophy by being executed locally on each AP when a probe request has been captured.
- 2) **Data processing:** For those cases where fingerprints might mask multiple devices with identical characteristics, a machine learning-based algorithm is implemented to differentiate them in the RSSI space. Initially, a pre-processing step aggregates and cleans the data. Subsequently, this processing technique is applied to groups of frames. These groups form clusters in an N -dimensional space, where N represents the number of APs that receive the frames. Typically, the effectiveness of this system improves as N increases, up to a certain limit.

A. FINGERPRINT GENERATION

In [12], the authors presented a stage where a stable anonymous identifier for Wi-Fi devices is acquired. As explained in Section II-A, in active scanning, the client station transmits

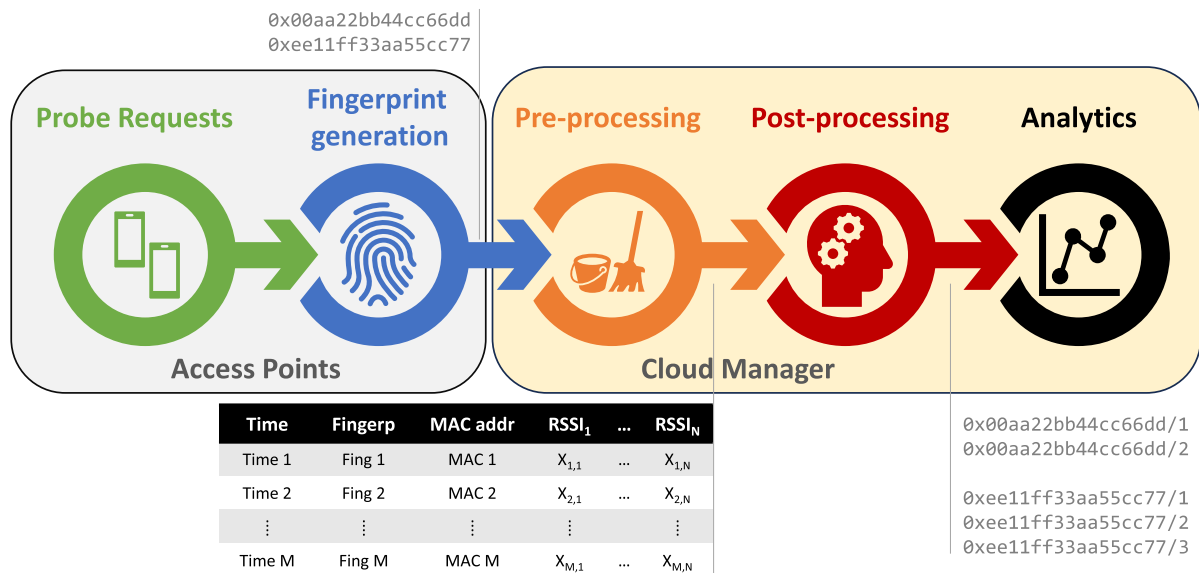


FIGURE 2. Overview of the whole system. First, Access Points receive the probe request frames and generate the fingerprint for each frame (Sec. IV-A). They send this information to the cloud infrastructure. There, the data is first pre-processed (Sec. IV-B) and post-processed (Sec. IV-C) to build the anonymous identifiers of the devices, with which Wi-Fi Analytics are inferred.

probe requests, and as with every 802.11 management frame, the probe request frame has a header and a body. The header contains the usual control and address fields, of which the only relevant field is the source MAC address, and the body comprises client station capabilities that can initially be compatible with an AP. A typical probe request frame format is shown in Table 1.

Therefore, this subsystem takes advantage of all the information present on each received probe request frame at the AP and locally generates a stable identifier or fingerprint using 5 GHz probe request frames. Frames in this band are preferred because they have more IEs, as shown by some ad hoc analyses that have been performed. For example, VHT-capabilities IE is rarely present in 2.4 GHz probe requests, as it corresponds to the IEEE 802.11ac standard, which operates only at 5 GHz. Therefore, at 5 GHz the entropy is higher and more information can be obtained with each probe request. The same process could be done for probe requests in the 2.4 GHz band and 6 GHz band. However, the former presents fewer IEs; therefore, the probability of two different devices sending the same body frame is higher. On the other hand, the 6 GHz band is not yet as widely used as the 5 GHz band, so many devices would go undetected. It is important to emphasize that it is necessary to select a single band for all fingerprinting, rather than combining different bands. This is because matching the probe request frames from the same device across different bands is unfeasible, even when they are emitted successively. It would be impossible to determine whether they originated from two distinct devices or from a single device.

This algorithm works for non-associated stations regardless of whether they use random MAC addresses. However, its role as a unique identifier for STAs is more significant when the MAC address is unreliable or randomized.

A fingerprint is created by applying a hash function to certain information contained within the frame, which has been adjusted using data obtained from tests in real environments. It is based on a simple principle: APs collect 802.11 probe request frames, extract their IEs, and then generate an identifier from those elements. Not all IEs present in a probe request frame are used; only those that are invariant for the same device and vary between different devices are used. In particular, the IEs used in the construction of fingerprints are:

- Supported rates.
- Extended supported rates.
- HT capabilities.
- VHT capabilities.
- HE capabilities.
- Extended capabilities.

Moreover, some residual MAC address information has been included ad hoc in the process, owing to the knowledge acquired from analyzing captures. In particular, the OUI of the MAC address is checked to determine whether it is a known fixed random prefix, such as Google's `da:a1:19`, as this provides additional information that can enhance the fingerprint because most current MAC addresses randomize 46 of the 48 bits.

Finally, all the obtained information, expressed as a chain of bits, is passed through a hashing algorithm, in particular, a 64-bit Fowler/Noll/Vo (FNV-1a) [36]. This hashing technique was selected because of its excellent balance among good distribution, low collisions, speed, simplicity, and portability. The resulting hash, for example, `0x570c2114f76974cc`, is the final fingerprint of the user device and condenses and cleans the information gathered on each probe request. A simple change in one IE results in a completely different character string owing to the hashing algorithm.

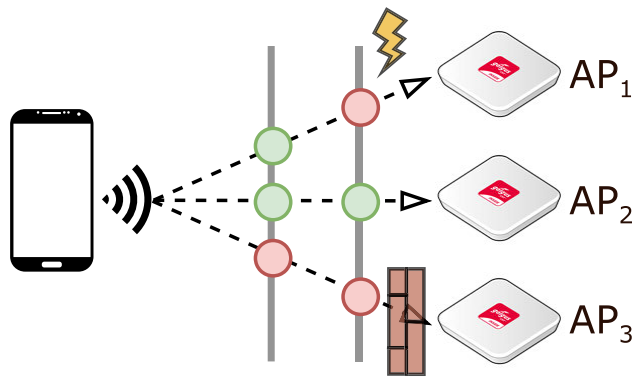


FIGURE 3. Example of possible behavior of 802.11 probe request frames in a real scenario. An STA sends two probe requests consecutively—each instant is depicted by a vertical grey line—. The first one is received only by AP₂, due to a collision in AP₁ and attenuation in AP₃. The second frame is successfully received by AP₁ and AP₂.

The fingerprint-generation algorithm runs locally on each AP. Galgus IC450 was chosen as the preferred model because of its versatility, which allowed us to modify any layer of its software. This capability ensured that we could tailor the router's functionality to meet the specific needs and requirements of the experiment. This process is executed for each received probe request, and the acquired information, along with the RSSI and timestamp of the reception is sent to the cloud infrastructure. It is important to highlight that APs must be synchronized, at least to the same tenth of a second, as the next stages use the sent timestamp. Each message is processed in real time as it is generated and stored in a database. Subsequently, a pre-processing stage is implemented to gather several almost simultaneous RSSI measurements for each captured fingerprint. Thus, it can be determined how the same frame is received by various APs.

B. PRE-PROCESSING OF RECEIVED PROBE REQUESTS

The pre-processing phase aims to identify which probe request frames have been captured by different APs, allowing for the RSSIs to be collectively analyzed in subsequent stages. This is an important step that must be considered because, if not properly treated, it can be a source of major problems and distortions. Consequently, two main steps were developed in this phase: burst identification and burst binding. A burst refers to a series of probe requests emitted by a device within a very brief time frame (on the order of milliseconds) intended to ensure data redundancy. During this short period, the MAC address remained unchanged. The number of probe requests sent per burst and how much time it waits to send the next one depend on the conditions of each device: driver version, OS version, battery level, and more.

In real scenarios, it is common for some frames in a burst to get lost while others successfully reach the AP because of collisions or possible interference. This situation is illustrated in Fig. 3. Two probe requests are emitted in every channel by the device, leading to three potential outcomes. For AP₁, one frame encounters a collision, indicated by a lightning symbol,

and is lost, whereas the other frame is received. For AP₂, both the requests are successfully received. However, for AP₃, an obstacle between the AP and the device causes attenuation and the consequent loss of both frames. Therefore, only AP₂ receives the first frame. If the probe request frames were associated individually, the data would first suggest that the device is far from AP₁ and AP₃, as none of them received the first frame, whereas the record of the second frame would indicate that the device is now closer to AP₁. This one-to-one probe request association would cause high noise in the measurements. To address this, the solution involves aggregating the received probe requests for each access point and burst.

The first step is to identify the different received bursts for each AP, grouping the probe requests frames received by the same AP with the same MAC address in the last second, as no device sends different bursts within that interval of time. These frames are collected together and both their RSSI values and timestamps are averaged. This procedure is known as burst identification.

Subsequently, burst binding is performed, where the measurements of the bursts received in different APs are associated. For this purpose, bursts are compiled and analyzed to identify instances where a burst with the same MAC address is received by different APs within a short time, typically a few seconds. If a burst with the same MAC address is detected at multiple access points within a short time frame, it is assumed to be from the same device. These bursts are then grouped in a structured format, such as a pivot table, for further analysis.

Finally, an optional step can be performed depending on the scenario in which the solution will be implemented. This is a filtering of the grouped bursts, discarding any burst where the highest received RSSI is below a specified threshold, typically around -80 dBm, to ensure that device detection is conducted in the vicinity of the APs, roughly controlled by the threshold value. For example, if a probe request burst is only received by one AP, and the maximum RSSI of the burst is -90 dBm, it can be assumed that the device is very far from the network in free-space propagation or that the frames have been highly attenuated owing to several obstacles. Therefore, depending on the scenario and the AP arrangement, it may be desirable to rule out these cases. As previously mentioned, the implementation of this filtering is optional and depends on the desired behavior. The resulting data structure is represented in Table 3.

C. POST-PROCESSING: UNSUPERVISED CLUSTERING

The purpose of this stage is to determine which of the acquired fingerprints conceals multiple devices and to segregate their probe request frames. This separation is essential for obtaining statistics and analytics such as tracking or device counting. The post-processing stage is conducted on the aggregated probe request bursts based on the time windows. The duration of these time windows is a design

TABLE 3. Structure of the pre-processed input of the algorithm. N represents the number of APs in the group, M is the number of captured bursts, K stands for the number of unique MACs, and F for the final number of unique fingerprints.

Time	Fingerprint	MAC	RSSI AP ₁	...	RSSI AP _N
Time 1	Fing 1	MAC 1	$x_{1,1}$	...	$x_{1,n}$
Time 2	Fing 2	MAC 2	$x_{2,1}$	...	$x_{2,n}$
...	...	...	...	...	...
Time M	Fing F	MAC K	$x_{m,1}$	...	$x_{m,n}$

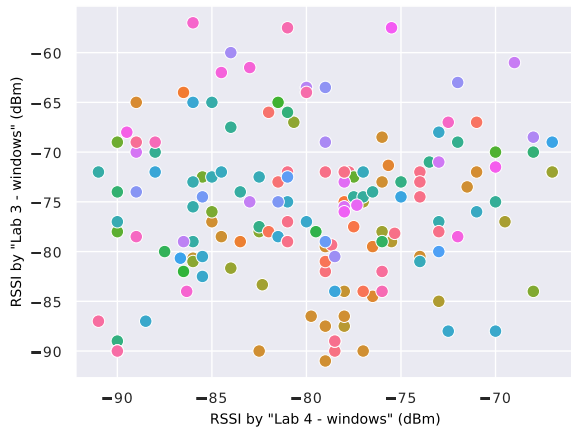


FIGURE 4. Scattering of RSSI of all the received probes with a randomized MAC within 5 minutes in a real environment. The vertical axis records the RSSI by AP “Lab3 - windows”, while the horizontal axis records the RSSI by AP “Lab4 - windows”. Each color represents a different MAC, with a total of 132 different MACs. No segmentation by fingerprint has been done yet.

variable that depends on the likely number of users in the vicinity.

Initially, within each window, the bursts are categorized based on the MAC address, whether real or randomized, by checking the (U/L) bit, as shown in Fig. 1. If it is a real MAC, the device is already unambiguously identified; therefore, analytics are trivial. However, as explained in Section II-B, most contemporary devices employ MAC address randomization. Consequently, a more comprehensive analysis is conducted. Fig. 4 displays all the probe request bursts with a random MAC received by only two APs in the site. A total of 132 different randomized MAC addresses were received during the 5-minute scanning time. It is clearly challenging to determine the number of devices that have generated all samples without segmentation based on fingerprints.

Therefore, the initial step is to group the bursts using their fingerprints. Subsequently, each group is scrutinized and clustered when necessary. The desired output is a list of labels or sub-fingerprints that identify which bursts belong to the same device. This determination is made by analyzing factors such as timing pattern, burst arrangement, and RSSI matrix. Hence, this stage can be summarized in two steps:

- 1) First, a temporal analysis is performed. The data are iterated to identify MAC addresses that appear many times within the time window. This implies

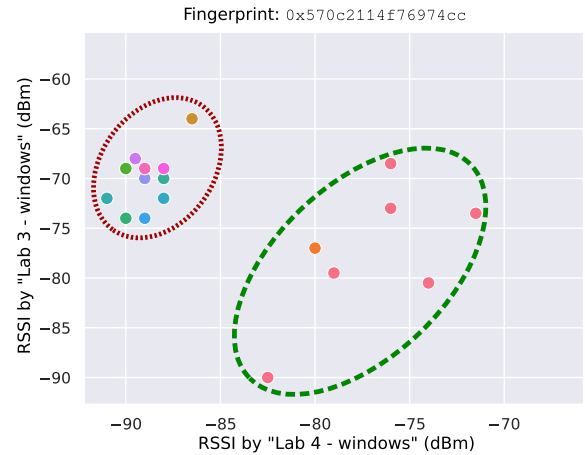


FIGURE 5. Representation of the clustering process, specified for fingerprint ‘0 × 570...4cc’, in a two-dimensional space. The color of the points stands for the MAC address of each processed probe request. Two different devices are said to be near the network with this fingerprint.

that they belong to a slow-rotation device, and only the bursts with that MAC belong to that device. Consequently, these bursts are assigned the same label in the subsequent step.

- 2) Unsupervised clustering is developed, and the algorithm assesses whether the resulting clusters are sufficiently compact. This assessment is conducted using a scoring metric that measures the maximum distance in the N -dimensional RSSI space between any two points within a cluster. Initially, it is determined whether a single cluster is adequate. If the spread score exceeds a certain threshold, the number of clusters is increased by one, and the clustering process is repeated. This loop continues until the probes are adequately concentrated within the resulting clusters, meaning that the spread score is lower than the threshold and the probes are labeled accordingly.

Fig. 5 shows the inputs and outputs of the clustering process. First, using points of different colors, the RSSIs of the received probe request frames for a particular fingerprint (0×570...4cc) are represented. It is evident at a glance that there are two distinct clusters, although, in some instances, such clear distinctions may not be immediately apparent, especially in higher-dimensional spaces. In addition, it is important to highlight that there are probes with the same MAC (light pink color) that are slightly dispersed because of the high mobility of the device and they are clustered together. Finally, the unsupervised clustering process is represented as two ellipses. It is ascertained that, within this time window, this same fingerprint concealed two different devices. A system that counts only one user per fingerprint would be incorrect.

Various clustering methods can be employed for this analysis, with k -means clustering being the most common. k -means is guaranteed to converge; however, initialization is critical for reaching the global minimum instead of a local minimum. Consequently, the clustering process is



FIGURE 6. Galgus access point model IC450.

executed multiple times with different initial centroids to enhance accuracy. An improvement to this implementation is the k -means++ scheme [37], which addresses initialization issues. k -means is a suitable choice for this application because of the similarity in the RSSI variance among pseudo-static devices in a given area. This is attributed to the short temporal span of the analysis and the high number of dimensions in the RSSI space. Alternatively, hierarchical clustering algorithms can be applied directly, and the number of clusters can be determined in a more straightforward manner.

V. EXPERIMENTAL TEST

A. MATERIALS AND METHODS

The experimental tests were conducted at the end of February 2023 in a classroom and laboratory building at the Universidad de Sevilla. Six Galgus IC450 APs (OS 11.2.2), shown in Fig. 6, which were previously deployed, were used to count the nearby devices using the proposed system. These APs were synchronized using a Network Time Protocol (NTP) server and connected to an extract-load-transform (ETL) cloud system based on Apache Kafka [38], where all probing events could be extracted and analyzed. Fig. 7 illustrates the deployment of the APs and the university campus and classrooms where the experiment was conducted. The six APs were interconnected using a Layer-2 switch, which was also connected to a router that provided access to the Internet. This is illustrated in Fig. 8. The APs locally generated the fingerprint for each received probe request and sent the probing event to Galgus ETL in real-time, each of which contains several key-value pairs, some of which being the timestamp of the reception of the frame, its generated fingerprint and its RSSI. The code that processes the events and generates the desired results was developed in Python, which uses the “kafka-python” library to create a consumer connected to the Apache Kafka broker. In this manner, the raw data are fed synchronously and subjected to the pre-processing and post-processing steps shown in Fig. 2. Fig. 9 shows a map of the laboratory classrooms where the experiments were performed. They were conducted in three adjoining classrooms, where several different classes were taught, with the consequent entrances and exits of students and peaks of affluence at class changes. For greater effectiveness and accuracy, six APs were used, two for each classroom, installed in the corners of the rooms, so that all



FIGURE 7. Scenario of the tests: the main campus building is shown at the top left; below, a laboratory classroom on a normal school day; on the right, Galgus technicians deploying the access points.

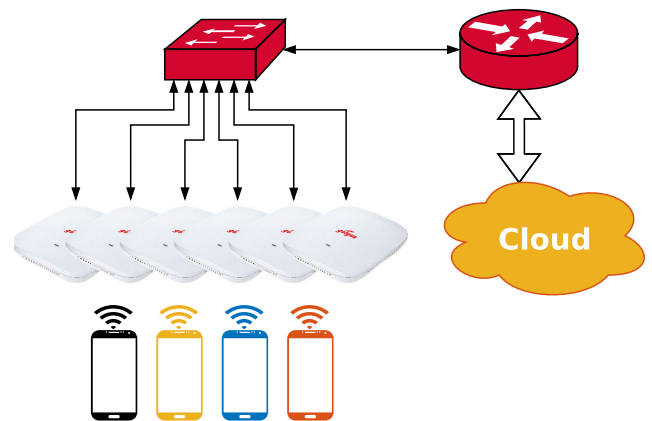


FIGURE 8. Block diagram of the experimental setup: the six APs that collect the frames are connected to a switch, which is linked to a router that give connection to the external network, hence, to the cloud.

the frames emitted by the devices of the people who were there could be captured. The entire test was analyzed in a time window of 30 min, and the maximum spread score for the clustering process was 22.

The experiment lasted more than six hours, and to compare the results of the system, the number of devices (smartphones, tablets, laptops, etc.) in the vicinity of the network was manually counted. It is crucial to note that individuals in the vicinity of the network were not active in the test. The participants were unaware of their involvement in the counting process. Their devices were not connected to the network but were merely present in the area: some devices were in use, others were idle, and some were stored away. The participants were oblivious to their participation in this study and all collected data were anonymized for privacy. As a key point, although 99-100% accuracy can be achieved in a controlled laboratory, this approach aimed to test the new system in a typical real-world environment with real users and their devices. In this context, the heterogeneity of devices and behaviors is very rich.

B. RESULTS

At the end of the experiment, 196 devices were counted by hand in total, with an average of 54.14 devices present in the building during each time window. This newly proposed

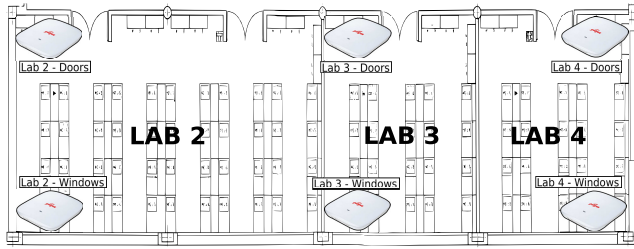


FIGURE 9. Map of the laboratory classrooms where the test took place and the location of the Access Points.

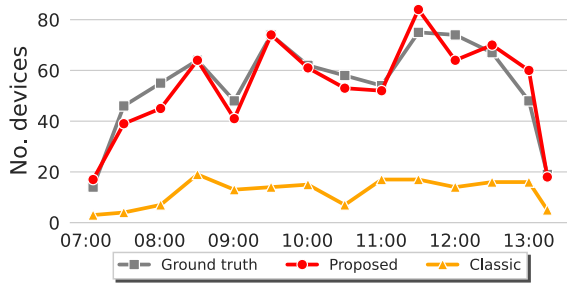


FIGURE 10. Comparison of counting of devices nearby the network every 30 minutes. Ground truth (count by hand) is in grey, while the count of the proposed system is shown in red, and the standard or classic industry approach is in yellow.

TABLE 4. Comparison of the attained metrics between the proposed system and the classic approach.

Metrics	Proposed system	Classic approach
MAE	5.0	42.2
RMSE	6.4	44.7
MASR	89.7%	22.2%
ACA	97.9%	22.0%

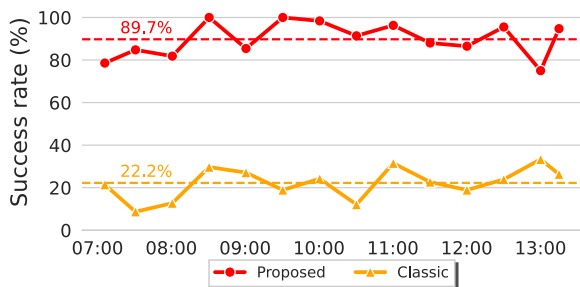


FIGURE 11. Comparison of the temporal success rate or accuracy percentage of the proposed system and the standard approach. The mean absolute success rate (MASR) of the proposed system is 89.7% while the standard approach reaches 22.2%.

de-randomization system counted 181 different devices, which has a total accuracy of 92.3%, while an average count of 53 devices was attained for each time window, which is an average counting accuracy (ACA) of 97.9% —53 over 54.14.

To emphasize the effectiveness of the proposed system, a comparison was performed with the classic and standard approach in the industry “count only real MACs” [9], [19], [20], [21]. The other methods mentioned in Section III can not be used for this comparison because they are outdated. In this experiment, the classic system counted only 40 devices in total —20.4% accuracy— with an average count of

11.9 devices —ACA of 22.0%, 11 over 54.14. Hence, it can be stated that the proposed system outperforms the current standard in terms of long-term convergence. Fig. 10 depicts the temporal counting of each system as well as the hand count as the reference in real-time. To clarify accuracy of each system, the success rate of each system for each time slot is shown in Fig. 11. It can be seen in the time series that the new system attained a mean absolute success rate (MASR) of 89.7%, with peaks of 100%. However, the classic Wi-Fi counting system reaches a maximum of 33% and an average success rate of 22.2%. Finally, error metrics, such as the mean absolute error (MAE) and root-mean-square error (RMSE), can be calculated. They are shown in Table 4 along with the ACA, and they encapsulate the improvements provided by this de-randomization system.

VI. CONCLUSION

All Wi-Fi analytics systems must face MAC randomization when attempting to obtain insights from non-associated STAs. Many applications have found this to be a key point; therefore, it is important to mitigate its effects. In this study, an analysis of the MAC randomization problem was conducted, as well as a study of state-of-the-art de-randomization techniques. Moreover, a new de-randomization system was proposed. A probe-request-based fingerprinting algorithm was developed and implemented for commodity Access Points. The APs run this algorithm and can segment the devices near the network with the resulting fingerprint. To improve the accuracy of the method, a machine-learning-based final stage was added to the segmentation using fingerprints. Fingerprinted probe request frames were collected by a central node or cloud server, and an unsupervised clustering algorithm is executed in the N -dimensional RSSI space. Hence, the distortion of Wi-Fi Analytics systems, such as the counting and tracking of non-associated STAs, was reduced. This study also presents an in-depth pilot test in a real-world 802.11 network. The results show that this system attains an MASR of almost 90% referring to the real-time counting, and an ACA of 98% referring to the average counting.

APPENDIX:

ACRONYMS AND ABBREVIATIONS

ACA	Average Counting Accuracy.
AP	Access Point.
HE	High efficiency.
HT	High throughput.
IE	Information element.
MAC	Media access control.
MAE	Mean absolute error.
MASR	Mean absolute success rate
NTP	Network time protocol.
OS	Operative system.
OUI	Organization Unique Identifier.
RMSE	Root-mean-square error.
RSSI	Received Signal Strength Indicator.

SSID	Service Set Identifier.
STA	Station, 802.11 user device.
UUID-E	Universally Unique Identifier-Enrollee.
VHT	Very high throughput.
WPS	Wi-Fi protected setup.

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